

Element I:

Test 1. Sun protection: Sun protection was the number one priority on our list because it is one of the project's requirements. We tested this by looking at the state of the plant. If it was dry, there was too much sun, if it was green, it was doing well. When we first tested the sun protection of our design, it wasn't very good, however, once we added the shade, the sun only hit the plant when we wanted it to. We tested the sun protection about five times. We pressed the button that made our platform spin, and most of the time the shade protected the plant, and when we wanted the sun to hit the plant it did. I would say the sun protection worked out about 4/5 times. The dependent variable was the amount of sun and the independent variable was time, or how long it spent in the sun. Overall, our sun protection tests were successful and the sun protection couldn't have gotten much better.

Test 2. Size: Size was our second highest priority because it was listed in the rubric of the project. We needed to have a 10 x 20 platform. We ended up doing a radius of 20 inches on a circular platform. We did not test the size when we were creating our platform, we used the rubric which said it had to be a 10x20. We ended up doing a circle platform with a radius of 20 inches. Our platform worked well and we didn't have any problems except trying to get the platform to spin. Overall the platform worked well in size and it worked well with our design.

Test 3. Support: Support was one of our medium priorities because everything that we created needs to have a stable foundation. Our requirements for support were to make sure it could hold ten pounds. The main ways we tested this were sitting our plant on top of our circular cut out and clicking the timer to see if it would rotate smoothly. We did way more than 10 trials because every time we tested to see if our engine was working, we were also testing the support as well. We had a bit of trouble at first because we needed more battery power to match the engine and our bottom was a little bit bendy, but we fixed it. In the end, we got our support to hold everything pretty well. One of our goals in the start for the support was to also make it pleasing to the eye, and I would say it does, and everything was very neat.

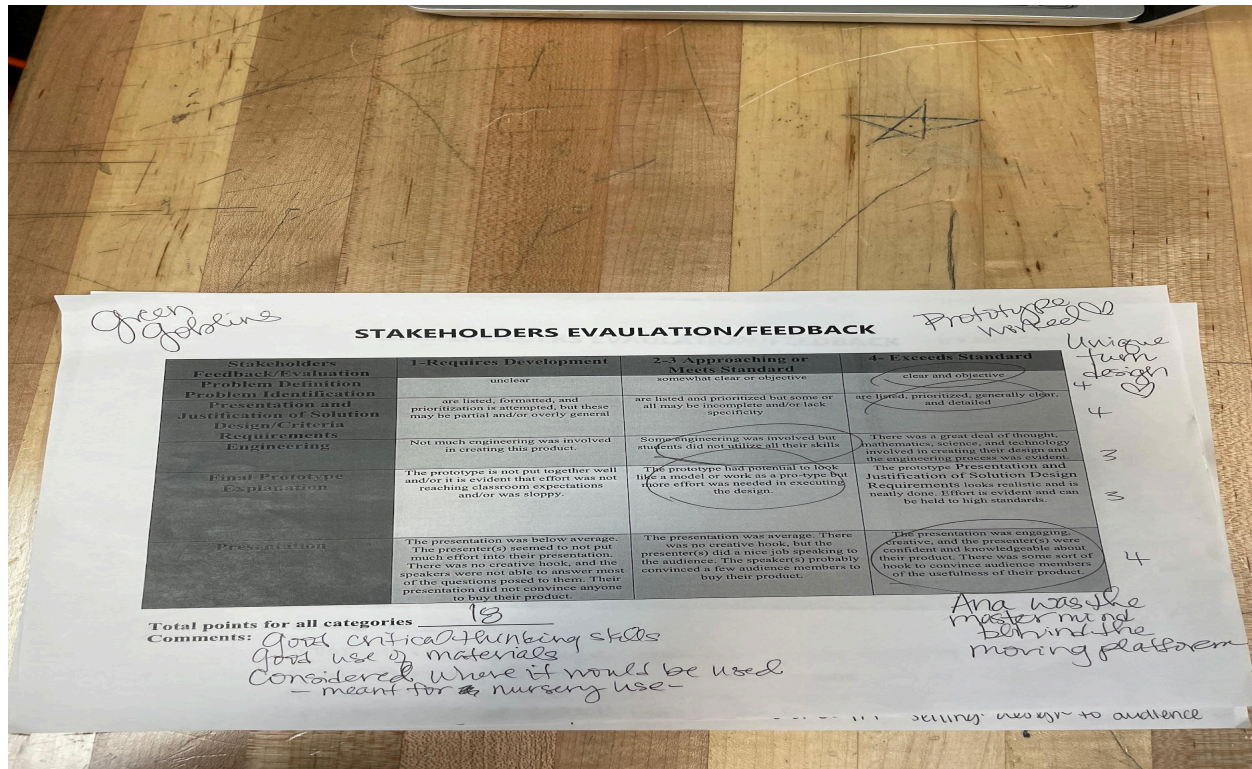
Test 4. Durability: This was the fourth priority, mostly because it was not in the criteria, but we still wanted it to be durable. We had to hit the shade ten times to see if it would stand up still. In our criteria we were supposed to make it useful for long periods of time and keep the plants protected. We had some issues with the bottom being a little bendy whenever we picked it up so we redid the bottom and put in a new piece of wood. A goal we had set for us was to make it indestructible and I would say we made it pretty good but there could be ways for it to break if we really wanted it too.

Test 5. Programmable: The programming part was not the most important, but it was still a needed piece of our project. Though we never were able to get to the programming test as we didn't have time, nor was anyone on the team able to program, as none of us know programming. The programming was tested around 20 times. We tested it so much because we struggled with getting the plant to stop in the sun. Each time it stopped somewhere different under the shade. It took us a while but we eventually got the plant to stop consistently under the sun and shade each time. Once we got the programming down our plant platform worked properly.

Test 6. Waterproof: This was one of our lowest priorities for this project because we didn't need our design to be waterproof; it wasn't in the criteria. We tried to make it waterproof but it just didn't work. The only thing on our project that needed to be waterproof was the batteries and wires. This was needed just to make sure the water doesn't mess up our battery powered things and water so nothing blows up or anything. We didn't really test it like we said we would because it was one of our lower priorities but we knew that it would work well enough. A goal we set before was to get 100% of the water to go to the plants and not our project but we never really got a chance to fully test that. This is one thing that we would definitely improve if we did this project again.

Test 7. Decoration: We had no requirements for decoration and it didn't matter to us if it was decorated. Even though it was our lowest priority, we still did some things like putting a turf and taping our wires together to also make it look a bit better. The decoration of our project was not tested. Decoration was not in the rubric so we did not worry about decoration. Therefore decoration was the lowest priority and we didn't worry about it.

Here are some comments and reviews from an expert who saw our project at the Design A Thon.



This picture shows our measurements that we used for our design.

